



Research article

Explainable AI-based Deep-SHAP for mapping the multivariate relationships between regional neuroimaging biomarkers and cognition

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ABSTRACT

Background: Mild cognitive impairment (MCI)/Alzheimer's disease (AD) is associated with cognitive decline beyond normal aging and linked to the alterations of brain volume quantified by magnetic resonance imaging (MRI) and amyloid-beta (A β) quantified by positron emission tomography (PET). Yet, the complex relationships between these regional imaging measures and cognition in MCI/AD remain unclear. Explainable artificial intelligence (AI) may uncover such relationships.

Method: We integrate the AI-based deep learning neural network and Shapley additive explanations (SHAP) approaches and introduce the Deep-SHAP method to investigate the multivariate relationships between regional imaging measures and cognition. After validating this approach on simulated data, we apply it to real experimental data from MCI/AD patients.

Results: Deep-SHAP significantly predicted cognition using simulated regional features and identified the ground-truth simulated regions as the most significant multivariate predictors. When applied to experimental MRI data, Deep-SHAP revealed that the insula, lateral occipital, medial frontal, temporal pole, and occipital fusiform gyrus are the primary contributors to global cognitive decline in MCI/AD. Furthermore, when applied to experimental amyloid Pittsburgh compound B (PiB)-PET data, Deep-SHAP identified the key brain regions for global cognitive decline in MCI/AD as the inferior temporal, parahippocampal, inferior frontal, supratemporal, and lateral frontal gray matter.

Conclusion: Deep-SHAP method uncovered the multivariate relationships between regional brain features and cognition, offering insights into the most critical modality-specific brain regions involved in MCI/AD mechanisms.

1. Introduction

AD is the leading cause of dementia, causing significant morbidity and mortality [1]. A β plaques accumulate in the brain, and early tracking of amyloid markers is crucial. Amyloid PET and MRI measurements are used to examine in-vivo amyloid and anatomical markers [2,3]. While recent clinical trials [4] show promising results in lowering amyloid burdens, their impact on neurodegeneration and cognition

remains minimal, indicating the complex and unresolved relationships among amyloid PET, MRI, and cognition.

Brain imaging methods, augmented by advanced AI algorithms [3,5], have become promising research pillars in mental diseases, aiding diagnosis, prevention, and drug discovery. Current brain-behavior studies have shown univariate relationships of individual brain regions with cognitive decline or clinical severity using univariate approaches [6], which only capture information from a single region at a

Abbreviations: MRI, Magnetic resonance imaging; PET, Positron emission tomography; PiB, Pittsburgh compound B; AI, Artificial intelligence; MCI, Mild cognitive impairment; AD, Alzheimer's disease; A β , Amyloid-beta; SHAP, Shapley additive explanations; MMSE, Mini-mental state examination; ROI, Region of interest; DVR, Distribution volume ratio; OASIS, Open access series of imaging studies; FDR, False discovery rate; MAE, Mean absolute error.

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time and are not designed to capture complex inter-relationships from multiple brain regions [7] collectively involved in cognitive processes. To address the limitation of univariate methods, we introduce the AI-based Deep-SHAP method and investigate multivariate relationships between regional imaging measures (amyloid PiB-PET and MRI) and cognition. The Deep-SHAP method was validated on simulated data and real data from MCI patients to test the hypothesis that (1) cognitive dysfunction can be estimated from regional A β and anatomical biomarkers, and (2) A β and anatomy-related dominant brain regions involved with cognition can be identified via the Deep-SHAP approach.

2. Materials and methods

2.1. Simulated and experimental data

The Open Access Series of Imaging Studies (OASIS) dataset [8] was

used, which included 173 controls for the simulation and 33 MCI participants for the experimental data. Written informed consent was obtained from all participants in this OASIS study following the institutional review board procedure at Washington University in St. Louis. The features include brain volumes assessed from the MRI or the distribution volume ratio (DVR) values derived from the PiB-PET, consistent with previous work [2]. The regions of interest (ROIs) were selected based on Braak staging regions [9] from previous literature in MCI/AD [10]. For simulating MCI/AD-related changes, we adjusted the brain volumes of the key regions and their corresponding MMSE in a fraction of subjects, resulting in a very strong correlation for these regions and weak correlations for others. The dataset and simulations are described in detail in the supplement methods S1, S2, and S3.

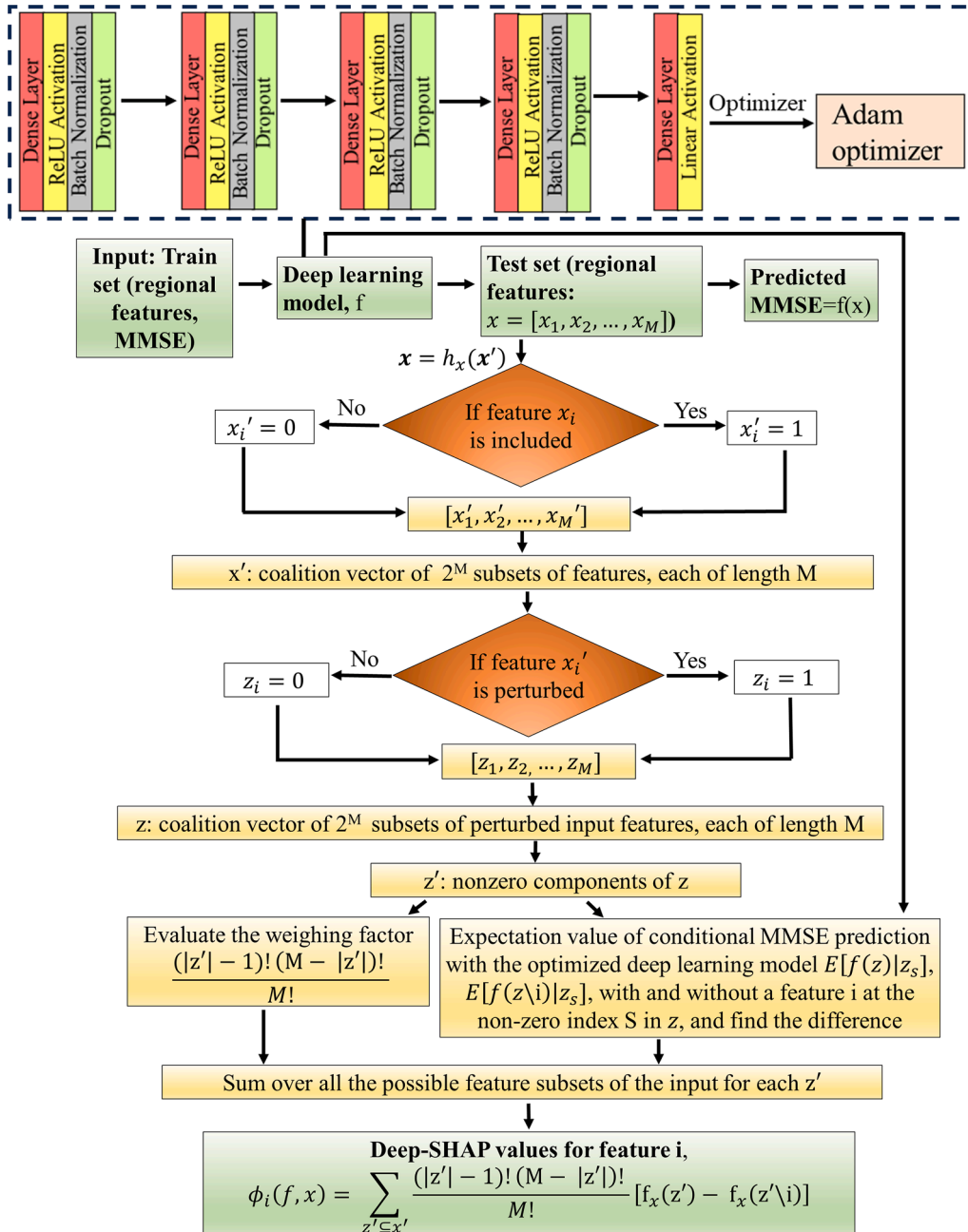


Fig. 1. Schematics of the Deep-SHAP method for estimating cognition and Deep-SHAP values (Deep learning model, f , is depicted within the dotted rectangular box).

2.2. AI-based Deep-SHAP method

We integrated the AI-based deep learning neural network (Deep) approach with the Shapley additive explanations (SHAP) [11] feature importance approach to introduce Deep-SHAP (Fig. 1). The model details are explained in supplement method S4.

Feature importance methods are useful because they help identify the most relevant features for a given problem, resulting in improved performance, comprehension, and more accurate predictions [5]. Furthermore, feature importance methods can shed light on the underlying relationships between the features and the target variable, which can be used to guide further analysis or domain-specific knowledge. To identify the contribution of each feature to the model's predictions, the SHAP [11,12] values were calculated for the test set in each fold of training via 10-fold cross validation [13]. SHAP comprises both model-agnostic as well as model-specific approximations that can be used to explain any black-box model. Herein, we are using the primary explainer interface of the SHAP library [11,14]. The primary foundation for the SHAP value is the Shapley value [15], a concept from cooperative game theory that assigns a fair value to each player in a game based on their contribution to the game's outcome. Let f represents a black box model that receives inputs, $x = [x_1, x_2, x_3, \dots, x_i, \dots, x_M]$, to train the model, and returns the output MMSE, indicated as y , where $y = f(x)$. Let x' be the simplified input that maps to the original input x by the mapping function $x = h_x(x')$. A coalition vector of length M , where each member is either 0 or 1, can be used to enumerate the simplified inputs $x' = [x'_1, x'_2, x'_3, \dots, x'_i, \dots, x'_M]$, which are all possible subsets of the features represented by x . There are 2^M possible subsets of features represented by x' . For instance, if x possesses three features (A, B, and C), then the features that could represent subsets of x' are $[0, 0, 0]$, $[0, 0, 1]$, $[0, 1, 0]$, $[1, 0, 0]$, $[0, 1, 1]$, $[1, 0, 1]$, $[1, 1, 0]$, and $[1, 1, 1]$. Every subset in x' matches a unique subset of features in x . z is a coalition vector that represents a perturbed version of the input features. Each element of z is either 0 or 1, depending on whether a feature in x' is included or excluded in the perturbed input. As an illustration, if x' is a subset containing the features A, B, and C $[1, 1, 1]$ and we only want to perturb the input of A and B, then z would be a vector $[1, 1, 0]$. The perturbed characteristics in the explanation model are represented by z' , which only contains the nonzero components of the binary vector z , i.e. $[1, 1]$ in our example. To determine the features that were modified, we examine z , which differentiates between altered and unaltered features. As a result, the expression for SHAP value can be written as:

$$\phi_i(f, x) = \sum_{z \subseteq x} \frac{(|z'| - 1)!(M - |z'|)!}{M!} [f_x(z') - f_x(z \setminus i)] \quad (1)$$

Here, $f_x(z') = f(h_x(z')) = E[f(z) | z_s]$ and $f_x(z \setminus i) = E[f(z \setminus i) | z_s]$. These expressions represent the expectation value of conditional MMSE prediction for the perturbed sample z' using the optimized deep learning model f , given a specific subset of predictors z_s , with the i^{th} feature included and excluded, respectively. Here, S denotes the collection of non-zero indexes in z , with the perturbed features and $|z'|$ is the cardinality of a subset z' of the input features x' . The SHAP value equation for feature i includes the term $(|z'| - 1)!(M - |z'|)!$ in the numerator of the weighing factor, representing the count of possible orders in which $|z'| - 1$ members of $z \setminus i$ can join the coalition before the inclusion of feature i and $(M - |z'|)$ members of z' can join after the inclusion of feature i . We are averaging over $M!$ permutations, and hence used as a normalization constant. This approach essentially handles the permutations of features within each coalition, without specifically focusing on the relative weight of individual features. Shapley values ensure equitable attribution by examining all possible feature orderings within each coalition. Even in cases of feature correlation, Shapley values consider the combined impact of features in the coalition, which may differ from the sum of their individual impacts. As a result, Shapley values inherently

capture interactions between features. In the context of a regression problem, the function $f(x)$ in equation (1) is substituted with a characteristic function v , which relies on the coefficient of multiple determinations (R^2 values) obtained from the regression model f with varying sets of features, as described in the equation, such that the sum of Shapley values over all the features equals the maximum value of R^2 with all the features together [16].

The SHAP approach is widely acknowledged for its robustness [17], but it is crucial to consider that executing SHAP with other approximate explainers might yield varying feature significance results due to the inherent randomness during data sampling. The SHAP should be used in such a way that the influence of random variability is reduced to improve the consistency of feature significance values. Additionally, because of variations in the model's weights and the impact of each feature on the model's predictions during each run, owing to the stochastic nature of neural network algorithms, the SHAP values may vary between different runs. We thus execute the Deep-SHAP model multiple times (four times in our case). We computed the average prediction for each participant over the four repeats to determine the correlation and p-value between the actual and predicted MMSE. Additionally, we calculated the absolute SHAP values for each feature across the dataset [18] and averaged these values over the four repeats to determine the relevance of each feature. We have also presented the training and test losses, employing MAE as the loss function, by computing the average across these four repetitions. This technique improves the consistency and reliability of feature significance estimations and enables us to explain which features served as the most significant predictors during the prediction of the target variable.

3. Results

3.1. Deep-SHAP validation in simulated MRI data

The Deep-SHAP model, when trained on simulated brain volumes derived from MRI data, yielded robust results. These results showed a substantial correlation of 0.92 with a highly significant FDR-corrected p-value of $1.04E-65$ between the true and predicted MMSE. Notably, the Deep-SHAP model demonstrated good generalization, as evidenced by a close match between the training loss (0.74) and the test loss (0.81). The features that exhibited the strongest correlations with MMSE, including the superior temporal gyrus, insula, supratemporal, opercular frontal, and occipital fusiform gyrus gray matter regions, not only emerged as the most influential but also emphasized the model's ability to accurately capture complex multivariate relationships. The brain slices showing the feature significance overlaid on a standard MRI template along with the correlation between the actual and predicted MMSE are shown in Fig. 2. In all figures, the brain slices are visualized based on axial orientation and correspond to slices numbered at positions $z = -28, -20, -14, 6, 0, 8, 16, 22, 30$, and 38 .

3.2. Deep-SHAP validation in simulated amyloid PET data

We further tested our Deep-SHAP model in simulated PET data, manipulating the MMSE scores and the DVR values of five selected brain features, including the lateral occipital, parietal, middle temporal, superior temporal, and cingulate gray matter regions. The model exhibited both training and test losses of 1.54 and 1.63, respectively, suggesting a balanced fit. Deep-SHAP approach yielded a robust and statistically significant correlation of 0.88, supported by an FDR-corrected p-value of $3.19E-52$. This analysis reiterated the manipulated features as the most influential ones, as found in the simulated MRI data. Fig. 3 displays the brain slices that exhibit feature significance overlaid on a standard MRI template, along with the correlation between the actual and predicted MMSE.

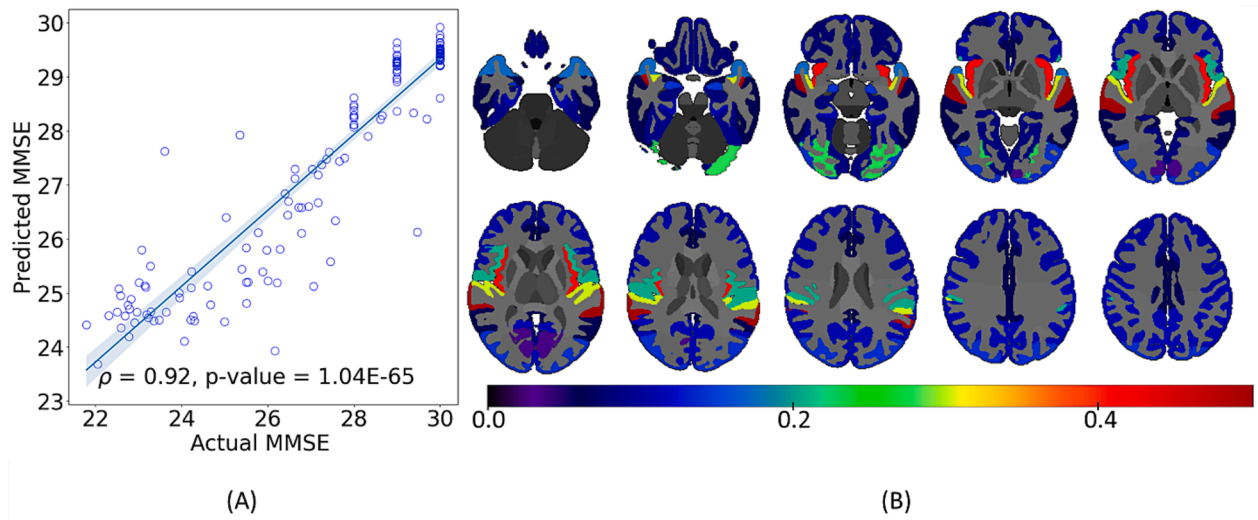


Fig. 2. Deep-SHAP on simulated MRI brain volumes. (A) illustrates the scatterplot of actual and predicted MMSE (solid blue line: linear fit, shadow: 95% confidence limit; ρ : Spearman correlation). (B) Brain slices showing significant brain regions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

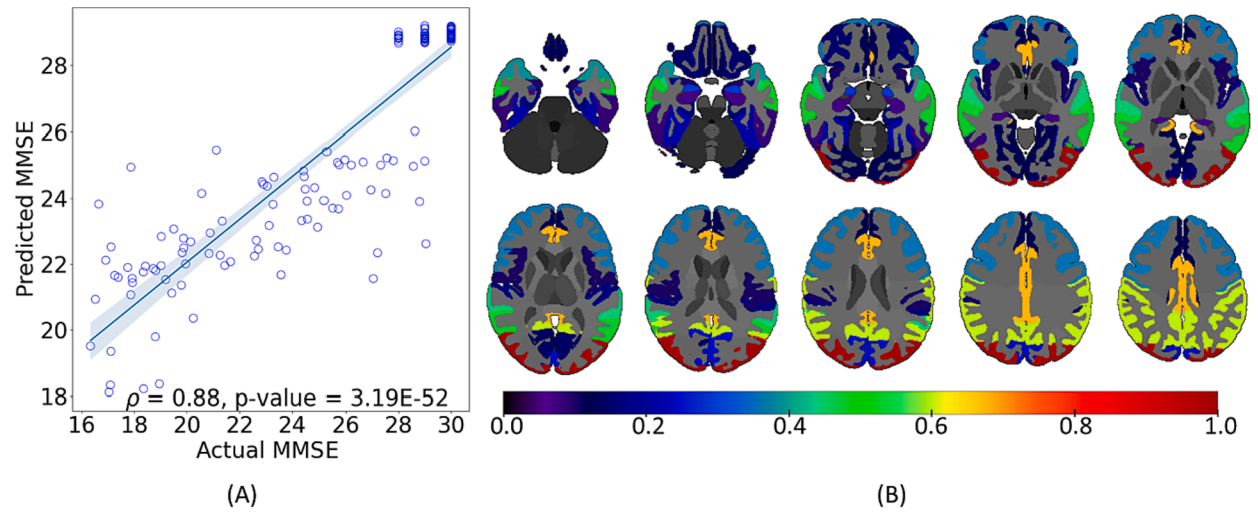


Fig. 3. Deep-SHAP on simulated PET DVR values. (A) The scatterplot of actual and predicted MMSE. (B) Brain slices showing significant brain regions.

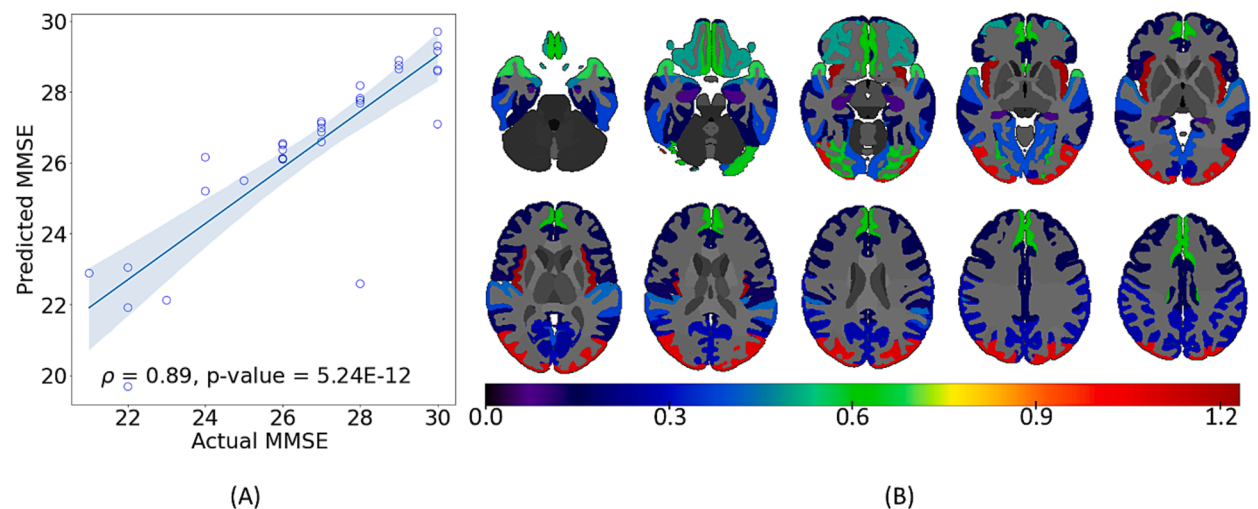


Fig. 4. Deep-SHAP on experimental MRI brain volumes. (A) Scatterplot of actual and predicted MMSE. (B) Brain slices showing significant brain regions.

3.3. Deep-SHAP application in experimental MRI data

When the Deep-SHAP model was employed on actual experimental MRI brain volumes, it revealed a robust and noteworthy correlation of 0.89 between actual and predicted MMSE, accompanied by an FDR-corrected p-value of $5.24\text{E-}12$. There was the well-fitted model exhibiting training and test losses of 0.93 and 0.96, respectively. Notably, the insula, lateral occipital, medial frontal, temporal pole, and occipital fusiform gray matter areas were identified as the most influential brain regions associated with MCI/AD. Fig. 4 displays the regions of significance overlaid onto a brain template as well as the scatterplot of the actual and predicted MMSE.

3.4. Deep-SHAP application in experimental amyloid PiB-PET data

The Deep-SHAP model was also applied to the real experimental amyloid PiB-PET data. It was trained using regional DVR values, resulting in a strong and statistically significant correlation of 0.93 between real and predicted MMSE, accompanied by an FDR-corrected p-value of $2.04\text{E-}15$. Likewise, the model displayed training and test losses of 0.77 and 0.78, respectively. The top five most influential features identified in this analysis were the inferior temporal, parahippocampal, inferior frontal, supratemporal, and lateral frontal regions. The significant brain regions superimposed on a brain template as well as the scatterplots between the actual and predicted MMSE are illustrated in Fig. 5.

4. Discussion

In this study, we introduced an AI-based Deep-SHAP method, integrating deep learning and SHAP. In simulations, Deep-SHAP significantly predicted cognition using regional imaging features and identified the ground-truth simulated regions as the most significant multivariate predictors. Using the Deep-SHAP method in MRI experimental data, we found the insula, lateral occipital, medial frontal, temporal pole, and occipital fusiform gyrus regions to be the most dominant contributors to the global cognitive decline processes in MCI/AD. Our Deep-SHAP approach, when applied to the experimental amyloid PiB-PET data, further yielded inferior temporal gyrus, parahippocampal gyrus, inferior frontal, supratemporal, and lateral frontal regions as the most significant predictors for global cognitive decline in MCI/AD.

Deep-SHAP results generalized well when compared to the related existing AI literature. Habuza et al. [19] employed a hybrid model that combined convolutional neural networks with linear regression using

structural MRI to predict MMSE and achieved an MAE of 2.38. Luo et al. [20] utilized machine learning algorithms incorporating brain cortical features, biological factors, and PET measurements to predict MMSE and achieved an MAE of 2.31. Lei et al. [21] employed various neural network models on MRI brain volumes and achieved an MAE of 1.56 for MMSE estimation. Comparing these findings, we observed our Deep-SHAP model's strong performance with the highest MAE of 1.63, highlighting its effectiveness in quantifying multivariate complex imaging patterns associated with cognitive decline in MCI/AD.

In MRI experimental data, Deep-SHAP findings align with prior research suggesting that regional atrophy [22], including alterations of the insula [23] and lateral occipital [24] regions might contribute to cognitive decline in MCI/AD. In experimental amyloid PiB-PET data, Deep-SHAP results were also broadly consistent with previous findings suggesting an association between the parahippocampal gyrus and cognitive functioning [25], significant molecular changes in its anterior sections [26], and corroborating evidence from various sources [27,28] during the MCI and early AD. Likewise, our Deep-SHAP model's predictions align with significant PiB retention in the inferior temporal gyrus among AD patients [28–30]. Furthermore, Laforce et al. [31] underscored the significance of the inferior frontal region in the context of MCI and early AD, aligning with our findings using Deep-SHAP.

This study revealed that MRI and A β PiB-PET markers have distinct dominant features. These markers may evolve through a structured progression rather than suddenly turning aberrant [32]. A β -plaque deposition typically plateaus in the early stages of AD, leading to detectable brain atrophy in MRI scans and cognitive symptoms [33]. Brain atrophy observed in imaging-based biomarkers could be due to neuronal death [34], neural degeneration [35], synaptic loss [29], and disruptions in axoplasmic flow [36]. Research supports the idea that brain regions implicated in MCI/AD may differ between these two biomarkers [37,38]. However, common ground in five brain regions from both modalities is found, highlighting the unique and overlapping aspects of MRI and PET in understanding AD mechanisms [39].

When dealing with a larger number of features, computing Shapley values becomes computationally demanding, involving the evaluation of 2^M subsets of features. As the number of features (M) increases, the computational complexity grows exponentially [11]. SHAP values are the conditional expectation function of the original model calculated by averaging over all possible permutations of features, as the feature contributions depend on the order in which they contribute. Hence, to alleviate the computation challenge, SHAP is approximated by combining insights from other feature attribution methods such as Kernel SHAP, Shapley sampling values [40], Linear SHAP, Tree SHAP, Max SHAP, Partition SHAP, Gradient SHAP [11,12], etc. Using SHAP,

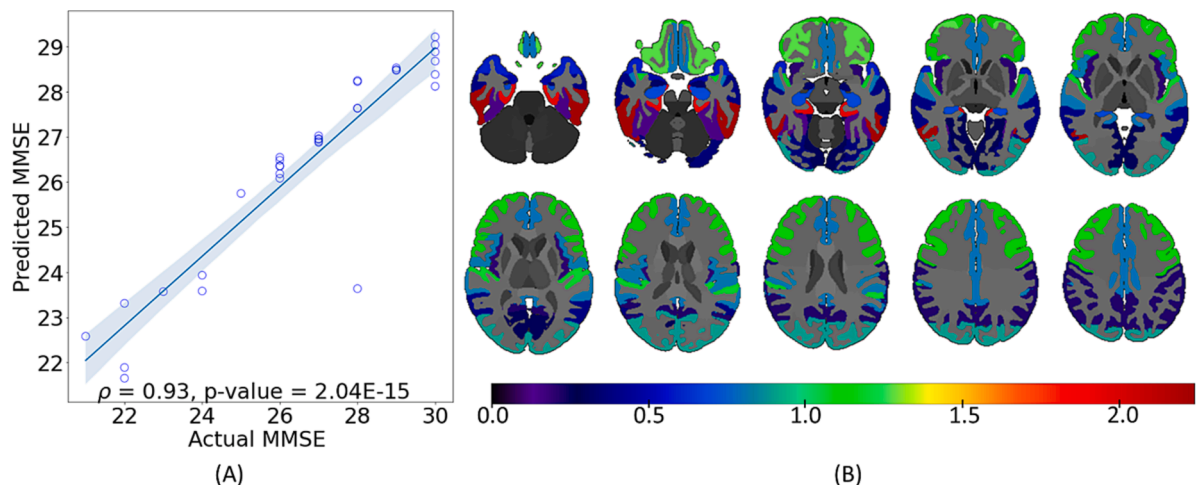


Fig. 5. Deep-SHAP on experimental amyloid PiB-PET DVRs. (A) The scatterplot of actual and predicted MMSE. (B) Brain slices showing significant regions.

global interpretations align with local explanations, as the Shapley values are the fundamental building blocks for the global interpretations [18].

In summary, we introduced the Deep-SHAP method and demonstrated its effectiveness in the simulated and real experimental MRI and amyloid PiB-PET datasets. In simulations, the Deep-SHAP method identified the ground-truth simulated regions as the most significant multivariate predictors for cognition. In the experimental data, the Deep-SHAP method comprehensively captured the modality-specific multivariate relationships between regional imaging features and cognition and demonstrated its potential to identify the most influential regional features collectively involved as network systems in predicting cognition in MCI/AD. These findings point to the potential use of AI-based diagnostic tools for identifying key hubs and developing targeted treatment strategies in the future. Future studies should focus on the potential applications of this method in a larger and more diverse sample size setting, offering additional insights into the complex multivariate brain vs. behavior relationships involved in MCI/AD mechanisms.

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CRediT authorship contribution statement

Puskar Bhattarai: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Deepa Singh Thakuri:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Yuzheng Nie:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Ganesh B. Chand:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejrad.2024.111403>.

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